

Introduction

SCHWESERNOTES - BOOK 2

There is a lot of overlap between the prerequisite materials and this reading. The prerequisites go into greater detail about the computation of tax bases for assets and liabilities, which may help with your understanding of this reading. The prerequisites also contain a numerical example of the impact of the enacted tax rate changing on existing deferred tax assets and liabilities.

Module 35.1: Differences Between Accounting Profit and Taxable Income

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LOS 35.a: Contrast accounting profit, taxable income, taxes payable, and income tax expense and temporary versus permanent differences between accounting profit and taxable income.

Financial accounting standards (IFRS and U.S. GAAP) are often different than income tax laws and regulations. As a result, the amount of income tax expense recognized in the income statement may differ from the actual taxes owed to the taxing authorities.

Let's begin by defining the terms we will use to distinguish items on a company's tax return (the information it files with its country's tax authorities) from the comparable items on its financial statements.

Tax Return Terminology

- **Taxable income.** This is income subject to tax based on the tax return.
- **Taxes payable.** This is the tax liability caused by *taxable income*. This is also known as the current tax expense, but do not confuse this with *income tax expense* (see next).
- **Income tax paid.** This is the actual cash flow for income taxes, including payments or refunds from other years.
- **Tax loss carryforward.** This is a current or past loss that can be used to reduce taxable income (thus, taxes payable) in the future. It can result in a deferred tax asset.
- **Tax base.** This is the net amount of an asset or liability used for tax reporting purposes.

Financial Reporting Terminology

- **Accounting profit.** This is pretax financial income based on financial accounting standards (also known as *income before tax* and *earnings before tax*).
- **Income tax expense.** This is the expense recognized in the income statement that includes taxes payable and *changes* in deferred tax liabilities and assets.
- **Deferred tax liabilities.** These are balance sheet amounts that result from an excess of income tax expense over taxes payable that are expected to result in future cash outflows.
- **Deferred tax assets.** These are balance sheet amounts that result from an excess of taxes payable over income tax expense that are expected to be recovered from future operations. These can also result from tax loss carryforwards.
- **Valuation allowance.** This is the reduction of deferred tax assets based on the likelihood the assets will not be realized.
- **Carrying value.** This is the net balance sheet value of an asset or liability.
- **Permanent difference.** This is a difference between taxable income (tax return) and pretax income (income statement) that will not reverse in the future.
- **Temporary difference.** This is a difference between the tax base and the carrying value of an asset or liability that will result in either taxable amounts or deductible amounts in the future. Several examples of how temporary differences arise are presented later in this review.

Differences Between Tax and Financial Reporting

Differences between the treatment of an accounting item for tax reporting and for financial reporting can occur when the following occur:

- The timing of revenue and expense recognition in the income statement and the tax return differ
- Certain revenues and expenses are recognized in the income statement, but never on the tax return or vice versa
- Assets and/or liabilities have different carrying amounts and tax bases
- Gain or loss recognition in the income statement that differs from the tax return
- Tax losses from prior periods may offset future taxable income
- Financial statement adjustments may not affect tax return, or may be recognized in different periods

Differences between the balance sheet carrying value and the tax base of an asset may be permanent or temporary. Temporary timing differences result from the same total amounts passing through the income statement and tax returns over time, but with differences in each individual period. These temporary timing differences result in the

creation of deferred tax assets and deferred tax liabilities, which we will discuss next. Permanent timing differences do not result in deferred tax assets or liabilities.

Deferred Tax Liabilities

A **deferred tax liability (DTL)** is created when the income tax expense (income statement) is greater than taxes payable (tax return) due to temporary differences. DTLs occur when

- revenues (or gains) are recognized in the income statement before they are included on the tax return due to temporary differences, and
- expenses (or losses) are tax deductible before they are recognized in the income statement.

DTLs are expected to reverse (i.e., they are caused by temporary differences) and result in future cash outflows when the taxes are paid. These may be referred to as "taxable temporary differences" because the firm will pay more tax when they reverse.

A DTL is most often created when an accelerated depreciation method is used on the tax return and straight-line depreciation is used on the income statement.

Example: Temporary timing differences

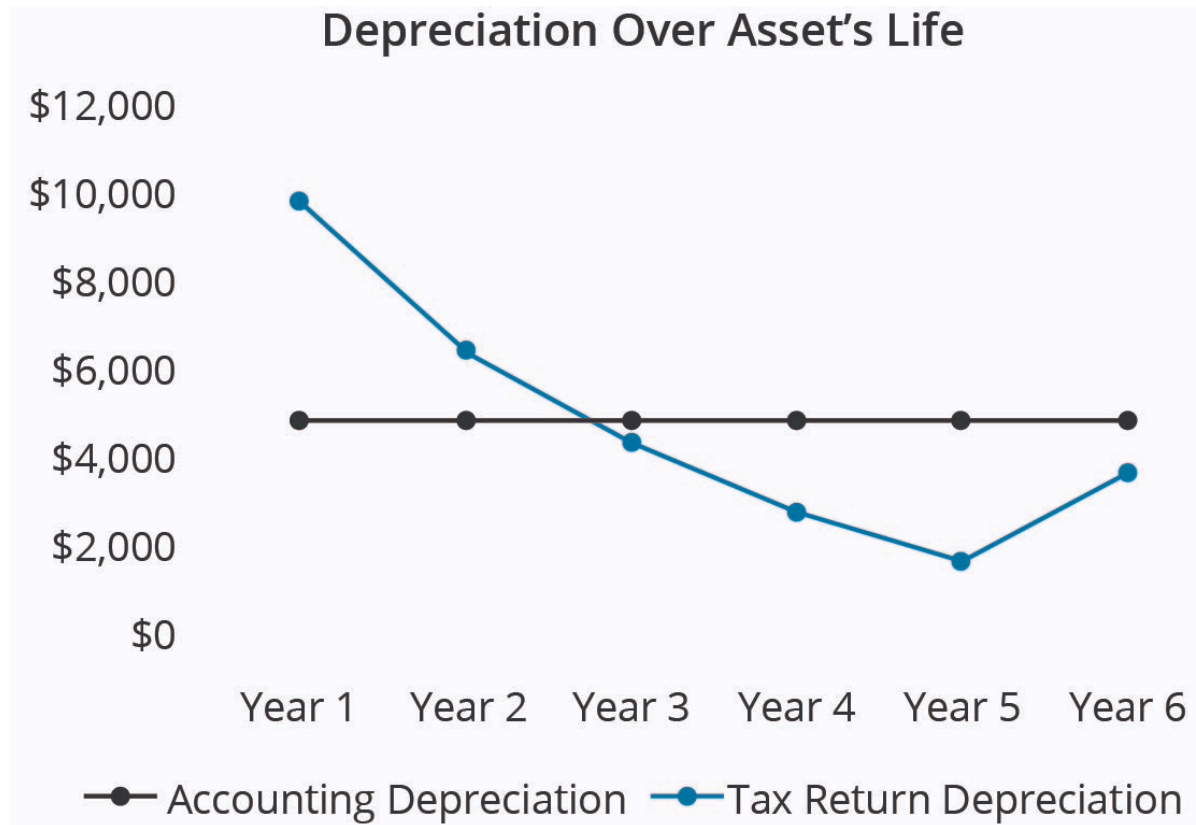
Flippy Discs Corporation purchased an injection-molding machine for \$30,000 at the start of the current period. The machine has a useful economic life of six years and no expected residual value. For accounting purposes, the machine will be depreciated straight line; however, a double declining balance will be used by the tax authorities. Calculate depreciation for accounting and for tax, and discuss the timing differences.

Answer:

[illegible]

Tax return depreciation	\$10,000	\$6,667	\$4,444	\$2,963	\$1,975	\$3,951	\$30,000
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The total amount of depreciation over the life of the asset is the same. The accelerated method used in the tax returns results in more depreciation in the early years of the asset's life, but less in the later years relative to the straight-line method. This is a classic example of a taxable temporary difference.



Visually, we can see that the tax return depreciation is greater in the first two years, creating a timing difference, which then begins to reverse from Year 3.

Due to the temporary timing differences, taxable income will initially be lower than earnings before tax in the accounts. The result of this is that Flippy pays lower tax in Years 1 and 2. Once the tax return

depreciation drops below the accounting depreciation, taxable income will be greater than earnings before tax, and Flippy will pay more tax. The temporary timing difference is created and then reverses, resulting in less tax due in the early years and more tax due in the later years. We know with certainty that when the timing difference reverses, Flippy will pay more tax, meeting the definition of a liability.

Deferred Tax Assets

A **deferred tax asset (DTA)** is created when taxes payable (tax return) are greater than income tax expense (income statement) due to temporary differences. DTAs occur when

- revenues (or gains) are taxable before they are recognized in the income statement,
- expenses (or losses) are recognized in the income statement before they are tax deductible, and
- tax loss carryforwards are available to reduce future taxable income.

Similar to DTLs, DTAs are expected to reverse through future operations. However, DTAs are expected to provide future tax savings, while DTLs are expected to result in future cash outflows. Timing differences that lead to the creation of DTAs may be referred to as "deductible temporary differences" because the creation or increase of a DTA will reduce the income statement tax expense. A firm that has taxable losses in excess of its taxable income can carry those excess losses forward and use them to reduce taxable income (and taxes) in future periods.

Post-employment benefits, unearned revenue, warranty expenses, and tax loss carryforwards are typical causes of DTAs.

Taxable and Deductible Temporary Differences

Initially, if taxable income (on the tax return) is lower than earnings before tax (on the income statement) because of a temporary timing difference, a DTL will be created. If taxable income is initially greater than earnings before tax, a DTA will be created.

Another way of analyzing temporary differences is to compare the balance sheet carrying values of assets and liabilities to their tax base values **Comparison of Carrying Value to Tax Base for Temporary Timing Differences.**

Comparison of Carrying Value to Tax Base for Temporary Timing Differences

Balance Sheet	Carrying Value vs. Tax Base	Deferred Tax Asset or Liability
Asset	Carrying value > tax base	Deferred tax liability
Asset	Carrying value < tax base	Deferred tax asset
Liability	Carrying value > tax base	Deferred tax asset
Liability	Carrying value < tax base	Deferred tax liability

PROFESSOR'S NOTE

The tax base of balance sheet assets and liabilities and the method of computation is covered in the prerequisite readings.

Permanent Differences

A **permanent difference** between taxable income and pretax income is one that will not reverse in the future. Permanent differences do not create DTAs or DTLs. Permanent differences can be caused by revenue that is not taxable, expenses that are not deductible, or tax credits that result in a direct reduction of taxes. For example, in the United States, interest received on municipal bonds is typically not taxable (but appears on the financial statements as pretax income), and the cost of life insurance on key company officers is typically not tax deductible (but appears on the financial statements as a pretax expense).

Permanent differences cause the firm's effective tax rate to differ from the statutory tax rate. We will examine statutory, effective, and cash tax rates later in this reading.

Tax Expense

A company's tax expense is the amount included in the income statement and may be referred to as a tax provision. It includes the amount the company owes the tax authorities (tax payable) based on this year's taxable income, along with any changes in DTAs and DTLs:

$$\text{tax expense} = \text{tax payable} + \Delta\text{DTL} - \Delta\text{DTA}$$

Creation of, or increases in, DTLs increase tax expense. Creation of, or increases in, DTAs reduce tax expense. This is because DTAs are expected to provide future tax savings, while DTLs are expected to result in future cash outflows for tax. Because the firm's actions in the current period either caused deferred tax items to be created or changed their values, the accruals concept requires the changes to be recognized in tax expense in the current period.

Changes in the Enacted Tax Rate

Changes in the company's statutory tax rate will also result in changes to DTLs and DTAs. Both DTLs and DTAs are created at the statutory tax rate in effect at the point of the initial timing difference. If the statutory tax rate changes, existing DTLs and DTAs must be adjusted. Because a DTL or DTA must reverse to zero when the timing differences reverse, they need to reflect the statutory tax rate expected to be in effect when reversal occurs. An increase in the statutory rate will increase both DTAs and DTLs, while a decrease will reduce them. Like any other change in a DTA or DTL, this will also affect tax expense in the income statement.

Module 35.2: Deferred Tax Assets and Liabilities

SCHWESERNOTES - BOOK 2

LOS 35.b: Explain how deferred tax liabilities and assets are created and the factors that determine how a company's deferred tax liabilities and assets should be treated for the purposes of financial analysis.

Creation of DTAs and DTLs

As discussed, DTAs and DTLs result from temporary timing differences between earnings before tax and taxable income. For a timing difference to be temporary, the amount that passes through the income statement and the tax return over the life of the asset or liability must be the same, even though the amounts differ that pass through each during individual periods.

Example: Deferred tax assets and liabilities

Firebird Corporation is a new subsidiary set up at the start of the year by its parent company, Wraith Incorporated. Firebird's statutory tax rate is 30%, and it has no permanent timing differences. Tax returns and income statements are presented for the first three years of its operation.

Tax return: Firebird Corporation

Year 1	Year 2	Year 3
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	\$000	\$000	\$000
Revenue	100,000	120,000	130,000
Cost of sales	28,500	34,200	37,050
Other expenses	19,950	23,940	25,935
Depreciation	10,000	10,000	10,000
Warranty costs	2,000	5,000	8,000
Interest expense	<u>10,000</u>	<u>12,000</u>	<u>12,500</u>
Taxable income	29,550	34,860	36,515
Tax payable 30%	8,865	10,458	10,955

Income statement: Firebird Corporation

	Year 1	Year 2	Year 3
	\$000	\$000	\$000
Revenue	95,000	114,000	123,500
Cost of sales	<u>28,500</u>	<u>34,200</u>	<u>37,050</u>
Gross profit	66,500	79,800	86,450
Other expenses	19,950	23,940	25,935
Depreciation	8,000	8,000	8,000
Interest expense	<u>10,000</u>	<u>12,000</u>	<u>12,500</u>
Earnings before tax	28,550	35,860	40,015
Tax expense	<u>8,565</u>	<u>10,758</u>	<u>12,005</u>
Net income	19,985	25,102	28,010

Firebird acquired PP&E for \$40 million at the start of Year 1. For financial statement reporting, the PP&E is depreciated straight-line over five years, while for tax reporting, it is depreciated straight-line over four years. Its residual value is assumed to be zero in both cases.

Firebird estimates that warranty expense will be 5% of gross revenues. For this example, we will assume all of the year-end liability values for warranties are likely to pass through future tax returns as the expenditure is incurred.

Analyze the differences between the tax return and the income statement, and calculate any deferred tax items that Firebird will recognize.

Answer:

The first noticeable difference is the revenue figures. Remember that revenue is reported net of any returns and allowances in the income statement (e.g., items such as estimated warranty provisions and customer discounts). The tax return has a line for warranty costs, which indicates that these may be the source of some of the difference. Another significant difference between the two statements are the depreciation lines. Finally, the tax return shows tax payable on taxable income, which differs from the tax expense in the income statement. The latter includes tax payable and changes in DTAs and DTLs.

1. Timing differences related to depreciation

$$\text{tax allowable depreciation} = \frac{\$40,000,000 - \$0}{4 \text{ years}} = \$10,000,000$$

$$\text{accounting depreciation} = \frac{\$40,000,000 - \$0}{5 \text{ years}} = \$8,000,000$$

In total, \$40 million of depreciation will pass through the tax returns and income statements, but different amounts will be reported for each individual year. Depreciation will be higher in the tax returns than in the income statement for each of the first four years. Firebird will recognize a DTL because the firm pays lower tax initially and higher tax on the reversal of the timing difference. We may also say the asset's carrying value is greater than its tax base, resulting in a DTL. After Year 4,

depreciation in the tax return will be zero, while depreciation in the income statement will continue, causing the timing difference to reverse.

End of Period	Year 1	Year 2	Year 3	Year 4	Year 5
	\$000	\$000	\$000	\$000	\$000
Carrying value	32,000	24,000	16,000	8,000	0
Tax base	<u>30,000</u>	<u>20,000</u>	<u>10,000</u>	<u>0</u>	<u>0</u>
Timing difference	2,000	4,000	6,000	8,000	0
DTL @ 30%	600	1,200	1,800	2,400	0
Δ DTL	+600	+600	+600	+600	-2,400

The timing difference grows over the first four years and then reverses in Year 5. Carrying value is equal to the asset's cost less accumulated accounting depreciation. The tax base it can be viewed as cost less accumulated tax-allowable depreciation. (Formally, it is defined as the total depreciation that will pass through the tax returns in future periods.)

2. Timing differences related to warranties

A warranty is a commitment to replace or repair a good if it becomes defective during the stated warranty period. Due to the accruals process— and in particular, the matching concept—the cost of

meeting future warranty claims must be estimated and accounted for at the time of the sale. The estimated future warranty expense is deducted from revenue, and a warranty liability is created in the balance sheet. Tax authorities, however, are unwilling to allow companies to save tax on estimated expenses. As a result, they typically require the expenditure to repair or replace faulty goods to have been incurred before they allow a deduction in the tax returns.

The following table shows the warranty provision, given Firebird's estimate of 5% of sales, and the expenditure actually incurred in each of the three years (from the tax return).

	Year 1	Year 2	Year 3
	\$000	\$000	\$000

Warranty provision	5,000	6,000	6,500
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Warranty expenditure	2,000	5,000	8,000
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The balance sheet liability will increase by each year's warranty provision, offset by the actual expense incurred:

Carrying Value	Year 1	Year 2	Year 3
	\$000	\$000	\$000

Warranty liability (at start of period)	0	3,000	4,000
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Warranty provision	5,000	6,000	6,500
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Warranty expenditure	(<u>2,000</u>)	(<u>5,000</u>)	(<u>8,000</u>)
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Warranty liability (at end of period)	3,000	4,000	2,500
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The tax base of a liability is its carrying value less the amount that *will* pass through future tax returns. (There is one exception to this, which is any form of revenue received in advance. These amounts are taxed when received—in other words, before income statement recognition—and the tax base becomes the carrying value of the liability less amounts that *will not* pass through tax returns in the future.)

Recall that we have assumed all of the year-end liability values for warranties are likely to pass through future tax returns as the expenditure is incurred. This will result in a zero value for the tax base in each year.

End of Period	Year 1	Year 2	Year 3
	\$000	\$000	\$000

Carrying value	3,000	4,000	2,500
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Tax base	<u>0</u>	<u>0</u>	<u>0</u>
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Timing difference	3,000	4,000	2,500
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DTA @ 30%	900	1,200	750
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Δ DTA	+900	+300	-450
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3. Income tax expense

Now that we have calculated the deferred tax items, we can illustrate how income tax expense was determined for financial statement reporting.

	Year 1	Year 2	Year 3
	\$000	\$000	\$000

Tax payable	8,865	10,458	10,955
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+Change in DTL	600	600	600
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-Change in DTA	<u>(900)</u>	<u>(300)</u>	<u>450</u>
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Tax expense	8,565	10,758	12,005
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Realizability of Deferred Tax Assets

Although deferred taxes are created from temporary differences that are expected to reverse in the future, neither DTAs nor DTLs are carried on the balance sheet at their discounted present value. However, DTAs are assessed at each balance sheet date to determine the likelihood that the company will have sufficient future taxable income to recover the tax assets. Without future taxable income, a DTA is worthless.

If there is sufficient doubt that the benefits of DTAs will be realized in future periods, both IFRS and U.S. GAAP require companies to recognize the diminished value of the DTAs. Under IFRS, companies simply reduce a DTA, which increases tax expense in that period. U.S. GAAP employs different mechanics with the same result. Under U.S. GAAP, the full DTA is still carried on the balance sheet, but it must be reduced by a **valuation allowance**. The valuation allowance is a contra account that reduces the net balance sheet value of the DTA. Creating or increasing a valuation allowance increases income tax expense and decreases net income.

It is up to management to defend the valuation of all DTAs. If a company has order backlogs or existing contracts that are expected to generate future taxable income, a valuation allowance might not be necessary. However, if a company has cumulative losses over the past few years or a history of inability to use tax loss carryforwards, then the company likely needs to reduce its DTAs to reflect the likelihood they may never be realized. Because of the subjective nature of estimating future profitability, adjustments to DTAs could be a source of potential earnings management.

Treatment for Analytical Purposes

If DTLs are expected to reverse in the future, an analyst should treat them as liabilities. If, however, an analyst does not expect them to reverse in the foreseeable future (typically because of expected continued growth in capital expenditures), the recommended adjustment is to treat them as equity (decrease the DTL and increase equity by the same amount). The key question is, "When or will the total DTL be reversed in the future?" In practice, the treatment of deferred taxes for analytical purposes varies. An analyst must decide on the appropriate treatment on a case-by-case basis.

Module 35.3: Tax Rates and Disclosures

SCHWESERNOTES - BOOK 2

LOS 35.c: Calculate, interpret, and contrast an issuer's effective tax rate, statutory tax rate, and cash tax rate.

Some firms' reported income tax expense differs from the amount that would result from multiplying its earnings before taxes by the statutory income tax rate. Recall that the statutory rate is the tax rate of the jurisdiction where the firm operates. Any differences generally result from the following:

- Different tax rates in different tax jurisdictions (countries)
- Permanent tax differences: tax credits, tax-exempt income, nondeductible expenses, and tax differences between capital gains and operating income
- Changes in tax rates and legislation
- Tax holidays in some countries

Typically, temporary timing differences do not cause the statutory and effective rates to differ.

Analysts should consider the following three tax rates:

1. **Statutory tax rate:** corporate income tax rate in which the company is domiciled
2. **Effective tax rate** =
$$\frac{\text{income tax expense}}{\text{pretax income}}$$
3. **Cash tax rate** =
$$\frac{\text{cash taxes paid}}{\text{pretax income}}$$

The footnote disclosure in the financial statements must reconcile the statutory tax rate to the effective tax rate. Understanding the differences between reported income tax expense and the amount based on the statutory income tax rate will enable the analyst to better forecast future earnings and cash flow.

When estimating future earnings and cash flows, the analyst should understand each element of the reconciliation, including its relative impact, how it has changed with time, and how it is likely to change in the future.

When analyzing trends in tax rates, it is important to only include reconciliation items that are continuous in nature rather than those that are sporadic. Items such as different tax rates in different countries, tax-exempt income, and nondeductible expenses tend to be continuous. Other items are almost always sporadic, such as large asset sales and tax holiday savings. Watch for special conditions such as termination dates for a tax holiday or a requirement to pay the accumulated taxes at some point in the future. An analyst should review the disclosures of each financial statement based on the footnotes and management discussion and analysis.

Example: Analyzing a tax rate reconciliation

Novelty Distribution Company (NDC) does business in the United States and abroad. NDC's reconciliation between effective and statutory tax rates for three years is provided in the following table. Analyze the changes in effective tax rates over the three years shown.

Statutory U.S. federal income tax rate reconciliation

	20X3	20X4	20X5
Statutory U.S. federal income tax rate	35.0%	35.0%	35.0%
State income taxes, net of related federal income tax benefit	2.1%	2.2%	2.3%
Benefits and taxes related to foreign operations	(6.5%)	(6.3%)	(2.7%)

Tax rate changes	0.0%	0.0%	(2.0%)
Capital gains on sale of assets	0.0%	(3.0%)	0.0%
Special items	(1.6%)	8.7%	2.5%
Other, net	0.8%	0.7%	(1.4%)
Effective income tax rates	29.8%	37.3%	33.7%

Answer:

In some cases an analyst may want to convert the reconciliation from percentages to absolute numbers. However, for this example, the trends can be analyzed simply by using the percentages.

The effective tax rate is upward trending over the three-year period. Contributing to the upward trend is an increase in the state income tax rate and a decrease in benefits related to taxes on foreign income. In 20X4, a loss related to the sale of assets partially offset an increase in taxes created by special items. In 20X3 and 20X5, the special items and the other items also offset each other. The fact that the special items and other items are so volatile over the three-year period suggests that it will be difficult for an analyst to forecast the effective tax rate for NDC for the foreseeable future without additional information. This volatility also reduces comparability with other firms.

LOS 35.d: Analyze disclosures relating to deferred tax items and the effective tax rate reconciliation and explain how information included in these disclosures affects a company's financial statements and financial ratios.

Companies are required to disclose details on the sources of the temporary differences that cause DTAs and DTLs to be reported on the balance sheet. Common examples of temporary differences include the following:

- A DTL results from using *accelerated depreciation for tax purposes* and straight-line depreciation for the financial statements. An analyst should consider the firm's growth rate and capital spending levels when determining whether the difference will actually reverse.
- *Impairments* generally result in a DTA because the write-down is recognized immediately in the income statement, but the deduction on the tax return is generally not allowed until the asset is sold or disposed of.
- *Restructuring* generates a DTA because the costs are recognized for financial reporting when the restructuring is announced, but they are not deducted for tax until actually paid. Restructuring usually results in significant cash outflows (net of the tax savings) in the years after the restructuring costs are reported.
- In the United States, firms that use LIFO for their financial statements are required to use LIFO for tax purposes, so no temporary differences result. However, in countries where this is not a requirement, temporary differences can result from the *choice of inventory cost flow method*.
- *Post-employment benefits* and *deferred compensation* are both recognized for financial reporting when earned by the employee, but they are not deducted for tax purposes until actually paid. These can result in a DTA that will be reversed when the benefits or compensation are paid.
- A deferred tax adjustment is made to stockholders' equity to reflect the future tax impact of unrealized gains or losses on *available-for-sale marketable securities* that are taken directly to equity. No DTL is added to the balance sheet for the future tax liability when gains or losses are realized.

Typically, the following deferred tax information is disclosed:

- DTLs, DTAs, any valuation allowance, and the net change in the valuation allowance over the period
- Any unrecognized DTL for undistributed earnings of subsidiaries and joint ventures
- Current-year tax effect of each type of temporary difference
- Components of income tax expense
- Reconciliation of reported income tax expense and the tax expense based on the statutory rate
- Tax loss carryforwards and credits

Example: Analyzing deferred tax item disclosures

WCCO, Inc.'s, income tax expense has consistently been larger than taxes payable over the last three years. WCCO disclosed in the footnotes to its 20X5 financial statements the major items recorded as DTAs and

DTLs (in millions of dollars), as shown in the following table.

Deferred tax disclosures

	20X5	20X4	20X3
Employee benefits	\$278	\$310	\$290
International tax loss carryforwards	<u>101</u>	<u>93</u>	<u>115</u>
Subtotal	379	403	405
Valuation allowance	<u>(24)</u>	<u>(57)</u>	<u>(64)</u>
Deferred tax assets	355	346	341
Property, plant, and equipment	452	361	320
Unrealized gains on available-for-sale securities	<u>67</u>	<u>44</u>	<u>23</u>
Deferred tax liabilities	519	405	343
Deferred income taxes	\$164	\$59	\$2

Given this information, explain why income tax expense has exceeded taxes payable over the last three years. Also explain the effect of the change in the valuation allowance on WCCO's earnings for 20X5.

Answer:

The company's DTA balance results from international tax loss carryforwards and employee benefits (most likely, pension and other post-retirement benefits), offset by a valuation allowance. The company's DTL balance results from property, plant, and equipment (most likely from using accelerated depreciation methods for tax and straight-line on the financial statements) and unrealized gains on securities classified as available for sale (because the unrealized gain is not taxable until realized).

Income tax expense is equal to taxes payable plus deferred income tax expense (change in DTLs minus change in DTAs). Because DTLs have been growing faster than DTAs, deferred income tax expense has been positive, resulting in income tax expense being higher than taxes payable.

Management decreased the valuation allowance by \$33 million in 20X5. This resulted in a reduction in deferred income tax expense and an increase in reported earnings for 20X5. Decreasing the valuation allowance implies that management has increased its estimate of future taxable income against which it can use DTAs.

Reading 35: Key Concepts

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LOS 35.a

Deferred tax terminology is as follows:

- *Taxable income*. Income subject to tax based on the tax return.
- *Accounting profit*. Pretax income from the income statement based on financial accounting standards.
- *Deferred tax assets (DTAs)*. Balance sheet asset value that results when taxes payable (tax return) are greater than income tax expense (income statement) and the difference is expected to reverse in future periods.
- *Deferred tax liabilities (DTLs)*. Balance sheet liability value that results when income tax expense (income statement) is greater than taxes payable (tax return) and the difference is expected to reverse in future periods.
- *Valuation allowance*. Reduction of DTAs (contra account) based on the likelihood that the future tax benefits will not be realized.
- *Taxes payable*. The tax liability from the tax return. Note that this term also refers to a liability that appears on the balance sheet for taxes due but not yet paid.
- *Income tax expense*. Expense recognized in the income statement that includes taxes payable and changes in DTAs and DTLs.

If taxable income is less than pretax income and the cause of the difference is expected to reverse in future years, a DTL is created. If taxable income is greater than pretax income and the difference is expected to reverse in future years, a DTA is created.

The balance of a DTA or DTL is equal to the difference between the tax base and the carrying value of the asset or liability, multiplied by the tax rate.

Income tax expense and taxes payable are related through the change in the DTA and the change in the DTL:

$$\text{income tax expense} = \text{taxes payable} + \Delta \text{DTL} - \Delta \text{DTA}$$

A permanent difference is a difference between taxable income and pretax income that will not reverse in the future. Permanent differences do not create DTAs or DTLs.

When a firm's enacted tax rate increases (decreases), DTAs and DTLs are both increased (decreased) to reflect the new rate. Changes in these values will also affect income tax expense.

LOS 35.b

A DTL is created when income tax expense (income statement) is higher than taxes payable (tax return). Deferred tax liabilities occur when revenues (or gains) are recognized in the income statement before they are taxable on the tax return, or expenses (or losses) are tax deductible before they are recognized in the income statement.

A DTA is created when taxes payable (tax return) are higher than income tax expense (income statement). DTAs are recorded when revenues (or gains) are taxable before they are recognized in the income statement, when expenses (or losses) are recognized in the income statement before they are tax deductible, or when tax loss carryforwards are available to reduce future taxable income.

The tax base for a depreciable fixed asset is the amount of tax-allowable depreciation that will be deducted in future tax returns.

The tax base of a liability is equal to the liability's carrying value less amounts that will be included in future taxable income. With liabilities relating to income received in advance, the tax base is equal to the carrying value of the liability less any amounts that will not be included in future taxable income.

Asset carrying value > tax base Deferred tax liability

Asset carrying value < tax base Deferred tax asset

Liability carrying value > tax base Deferred tax asset

Liability carrying value < tax base Deferred tax liability

If it is more likely than not that some or all of a DTA will not be realized (because of insufficient future taxable income to recover the tax asset), then the gross DTA must be reduced by a valuation allowance under U.S. GAAP. Under IFRS a firm reports a smaller DTA if future recoverability is uncertain but does not report a valuation allowance.

DTLs that are not expected to reverse, typically because of expected continued growth in capital expenditures, should be treated for analytical purposes as equity. If DTLs are expected to reverse, they should be treated for analytical purposes as liabilities.

LOS 35.c

Firms are required to reconcile their effective income tax rate with the applicable statutory rate in the country where the business is domiciled. Analyzing trends in individual reconciliation items can aid in understanding past earnings trends and in predicting future effective tax rates. Where adequate data are provided, they can also be helpful in predicting future earnings and cash flows, or for adjusting financial ratios.

Tax rates are as follows:

1. Statutory tax rate: corporate income tax rate in which the company is domiciled
2. Effective tax rate =
$$\frac{\text{income tax expense}}{\text{pretax income}}$$
3. Cash tax rate =
$$\frac{\text{cash taxes paid}}{\text{pretax income}}$$

LOS 35.d

Typically, the following deferred tax information is disclosed in the footnotes:

- DTLs, DTAs, any valuation allowance, and the net change in the valuation allowance over the period
- Any unrecognized DTL for undistributed earnings of subsidiaries and joint ventures
- Current-year tax effect of each type of temporary difference
- Components of income tax expense
- Reconciliation of reported income tax expense and the tax expense based on the statutory rate
- Tax loss carryforwards and credits